Dipesh Tamboli

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Research Scientist working on alignment and trustworthy AI for LLMs and foundation models. My research spans multi-objective preference optimization (DPO/RLHF), test-time alignment, and privacy-preserving generative AI, with first-author publications at ICML, ACL, and IEEE Transactions on AI, and rigorous evaluation grounding every method. I translate alignment research into production systems serving millions of users. ICML 2026 Gold Reviewer and CVPR 2025 Outstanding Reviewer; 262 citations, h-index 9.

Experience

Research Scientist — *Menlo Park, CA*
Meta Platforms, Inc. (Agent Data and Optimization, Applied AI) — *Jun 2025 – Present*

- Research on multi-objective alignment, privacy-preserving training, and trustworthiness evaluation for large language and foundation models, translating alignment methods into production systems.
- Built an LLM-as-judge system by distilling a frontier model (Gemini 2.5 Pro) into a production judge for content-quality and safety signals, preserving judgment fidelity while cutting per-task inference cost by ~$5M annually; reused across 5+ tasks.
- Co-authored research on latent reasoning and System 1 / System 2 communication in LLMs (NeurIPS 2025 workshop).
- Led a production ranking-model refresh validated by rigorous offline and online experiments (**+9% Click AUC**, **+28% normalized entropy**), serving millions of users.

Doctoral Researcher, Generative AI Alignment & Evaluation — *West Lafayette, IN*
Purdue University (Advisor: Prof. Vaneet Aggarwal) — *Aug 2021 – Aug 2025*

- Led an independent research agenda on LLM and generative-model alignment and evaluation, producing first-author publications at **ICML, ACL, TMLR, and IEEE Transactions on AI**; PRIVATEEDIT yielded a U.S. provisional patent and coverage by 21 outlets.
- Developed multi-objective preference optimization (BalancedDPO) and test-time alignment (TARo), benchmarked across reasoning, instruction-following, and generation tasks against strong baselines.
- Concurrent research internships: **Intern**, Meta (2024); **Applied Scientist Intern**, Amazon Robotics (2023).

Selected Publications

(† = first author; full list on Google Scholar)

- † **BalancedDPO: Adaptive Multi-Metric Alignment.** *TMLR*, 2026. Multi-objective preference optimization (DPO) aligning models to several objectives at once without reward hacking.
- **TARo: Token-level Adaptive Routing for LLM Test-time Alignment.** *ACL Findings*, 2026. Reward-model-guided test-time alignment that steers a frozen LLM toward structured reasoning without retraining.
- † **PRIVATEEDIT: A Privacy-Preserving Pipeline for Face-Centric Generative Image Editing.** *IEEE Transactions on AI*, 2026. Privacy-by-design generative AI; covered by 21 outlets; U.S. provisional patent.
- **Exploring System 1 and System 2 Communication for Latent Reasoning in LLMs.** *NeurIPS 2025 Workshop (FoRLM).*
- † **PosNegDM: Reinforced Sequential Decision-Making for Sepsis Treatment.** *IEEE J. Biomedical and Health Informatics*, 2024. Reinforcement learning with reward decomposition.
- **Multi-task Hierarchical Adversarial Inverse Reinforcement Learning.** *ICML*, 2023.

Awards & Professional Service

- **ICML 2026 Gold Reviewer Award** (top-reviewer recognition on area-chair ratings); **CVPR 2025 Outstanding Reviewer**; ECCV 2024 and ICML 2026 Emergency Reviewer.
- **39 peer reviews across 14 venues** (CVPR, ICML, ICCV, ECCV, AAAI, WACV, and journals), evaluating research quality and methodological rigor.
- **262 citations, h-index 9, i10-index 9** (Google Scholar); Best Paper Award, IEEE WIECON-ECE 2019.

Education

- **Ph.D., Electrical and Computer Engineering**, Purdue University, 2025.
- **B.Tech., Electrical Engineering** (Minor in Computer Science), IIT Bombay, 2021.

Technical Skills

Languages: Python, C++, SQL. **Frameworks:** PyTorch, TensorFlow. **Focus:** LLM alignment and trustworthy AI, post-training (DPO/RLHF), test-time alignment, privacy-preserving ML, reinforcement learning, evaluation and benchmarking, large-scale inference.